

SOUMYADEEP ROY CHOWDHURY

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PROFILE

Tech enthusiast with hands-on experience architecting highly concurrent distributed systems, AI-integrated software. Passionate about tackling complex data structures, building scalable, high-performance web applications.

EDUCATION

Jadavpur University, Kolkata - CGPA : 8.83 (till 3rd Semester) Expected 2028
B.E. In Information Technology

Vivekananda Mission School, Joka - 99.5 % (ISC AIR-2) 2024
Higher Secondary Examinations [Class - XII]

Vivekananda Mission School, Joka - 98.6 % (ICSE) 2022
Secondary Examinations [Class - X]

SKILLS

- **Relevant Coursework** — OOP, DSA, DBMS, CN, System Design
- **Languages** — C, C++, Java, Python, JavaScript, Solidity, SQL
- **Frameworks & Libraries** — React.js, Next.js, Tailwind CSS, Express.js, Langchain
- **Tools & Technologies** — Git, Docker, AWS, Google Generative AI, Jupyter
- **Databases** — MySQL, MongoDB, Redis, FAISS

PROJECTS

RAG-CRAW : [\[Demo\]](#) [\[Github\]](#) *Langchain | Selenium | FAISS | Google Gen AI |* [\[Video\]](#)
- Engineered a recursive Python web crawler to autonomously extract and process dynamic HTML content.
- Architected a LangChain RAG pipeline using HuggingFace embeddings to enable rapid semantic search.
- Automated data ingestion and document chunking with the Unstructured library to format raw web data.
- Built an interactive Streamlit UI integrated with Google Generative AI, allowing users to query scraped sites using natural language.

QUIZZY : [\[Demo\]](#) [\[Github\]](#) *React.js | Redis | Socket.io | Docker |* [\[Video\]](#)
- Built an event-driven microservices platform, to eliminate HTTP polling and achieve sub-millisecond data sync.
- Optimized high-concurrency leaderboards using Redis Sorted Sets and a composite scoring algorithm.
- Containerized monorepo workflows, configuring legacy watch-polling for seamless WSL2 file synchronization.
- Scaled a stateless cloud architecture to support concurrent users by mitigating event loop starvation.

GREENLENS : [\[Demo\]](#) [\[Github\]](#) *Next.js | Ethers.js | Cloudinary | Clerk & JWT |* [\[Video\]](#)
- Built a MERN-based dApp that tracks carbon footprints and rewards users with ERC-20 GreenTokens.
- Integrated a Python/Flask regression model to calculate real-time CO2 emissions from user utility.
- Deployed Solidity contracts on the Sepolia testnet to automate transparent, decentralized reward minting.
- Created a React/Tailwind UI featuring ethers.js wallet integration and dynamic data visualization charts.

ACHIEVEMENTS

- Ranked 46 & 146 in ICPC Asia West [Chennai](#) and [Amritapuri](#) Regionals 2025 respectively.
- Reached Specialist in [CODEFORCES](#) with a Maximum Rating of 1448 and 3 Star on [CODECHEF](#).
- Solved over 500+ problems across all competitive programming platforms.
- Finalist of Eastern India Blockchain Summit 2.0 at IIT Kharagpur.
- Achieved 98%ile in JEE Mains 2024 a rank of 326 in WBJEE 2024.

CO-CURRICULAR ACTIVITY

ACM Jadavpur University Student Chapter - Core Member of Web3 Team